

Marking Guide (Instructor Use)

Final Take-Home Assignment — Clinical Data Analysis in R, Phase I

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Total: 100 marks

This guide gives a weighted rubric, an answer key with canonical results, a common-errors penalty list, and grade bands. Mark on **correct method and sound interpretation**, not on exact second-decimal matches — package and R versions shift decimals trivially. Reward justified, well-documented decisions even where they differ slightly from the model answer.

Weighting summary

#	Component	Marks
1	Data import & cleaning	20
2	Descriptive statistics & Table 1	15
3	Figures	10
4	Statistical tests	15
5	Regression modelling	20
6	Interpretation & Results writing	15
7	Reproducibility & code quality	5
	Total	100

Component rubrics

1. Data import & cleaning — 20 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Import raw file & inspect	3	Reads the raw csv/xlsx; inspects structure/dims; notes columns mis-typed by sentinels	Imports but no inspection / no comment on types	Starts from a cleaned file, or import fails
Remove 3 duplicates → 1,500	3	Drops duplicates, confirms 1,500 unique patients, reports count removed	De-duplicates but doesn't verify/report	Duplicates left in (1,503 rows)
Missing-value sentinels → NA	4	All of blank/NA/999/-99 converted to NA in the right columns	Some sentinels handled, others missed	Sentinels left as numbers (e.g. 999 in glucose)

Sub-criterion	Marks	Full credit	Partial	Zero
Inconsistent categories standardised	4	sex, all mixed binaries, and whitespace fields harmonised consistently	Partial harmonisation; residual variants remain	No recoding; F/Male/1/0 left mixed
Impossible values screened	3	Identifies & justifiably handles age 0/200, SBP 0/700, DBP 5, weight 7, height 17	Handles some; rule unstated	Implausible values left in analysis
Dates parsed	1	enroll_date parsed to Date across all 3 formats	Partial parsing	Not parsed
Recode/derive + factor reference levels	2	BMI recomputed; sensible reference levels; education/PA ordered	Some done	None

Earns marks: `na_if()` / `case_when()` used cleanly; counts reported at each step; a documented rule ("values outside physiological range set to NA").

Loses marks: dropping whole rows with any NA before cleaning; "fixing" by guessing values without justification; trusting the supplied `bmi`.

2. Descriptive statistics & Table 1 — 15 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Appropriate summaries by type	4	Mean±SD or median[IQR] for numeric; n(%) for categorical	Mixed/inappropriate choices	Raw dumps only
Table 1 stratified by <code>treatment_uptake</code>	5	Clean, labelled table stratified by outcome	Built but not stratified, or unlabelled	No Table 1
Comparison test column	3	Sensible per-variable p-values included	Some tests wrong/missing	None
Publication quality & export	3	Readable units/labels; exported (docx/html/png)	Rough formatting	Not exported

Earns marks: `gtsummary::tbl_summary()` | `> add_p()` (or `flextable`) with clear labels and units.

Loses marks: Table 1 built on all 1,500 instead of the cleaned cohort; percentages that don't account for missing; unrounded values (e.g. 47.382%).

3. Figures — 10 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Two relevant figures	4	≥2 figures that address the question	Only one figure	None
Quality (titles, axis labels, units, legibility)	3	Fully labelled, legible, sensible chart type	Missing labels/units	Unlabelled
Exported at 300 dpi	3	<code>ggsave(..., dpi = 300)</code> (or equivalent) evidenced	Exported but dpi not set	Not exported

Earns marks: forest plot of aORs; uptake-by-determinant bar chart with counts/%; boxplot of knowledge score by uptake.

Loses marks: default ggplot with no labels; screenshots instead of exported files; 3-D / pie charts.

4. Statistical tests — 15 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Restrict to diagnosed (<code>htn_diagnosed=="Yes"</code>)	4	Filters to ~ 1,089 ; reports n	Filters but doesn't report	Tests run on all 1,500
Correct test choice	5	Chi-sq/Fisher for categorical; t/Wilcoxon for numeric; assumptions considered	Some mismatches	Wrong tests throughout
Correct execution & reporting	4	Statistics + p-values reported clearly	Incomplete reporting	Not reported
Sensible variable screening	2	Candidate determinants tested	Ad hoc selection	None

Earns marks: Fisher used where a cell is sparse (e.g. diabetes × uptake); Wilcoxon when distribution is skewed.

Loses marks: chi-square on tiny expected counts without comment; using the full sample; p-values with no test named.

5. Regression modelling — 20 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Univariable logistic models	4	Crude ORs + 95% CIs per predictor	ORs without CIs	Not done
Multivariable model specification	5	<code>glm(..., family=binomial)</code> with the pre-specified determinants, on diagnosed cohort	Some predictors missing/mis-specified	Wrong outcome/population

Sub-criterion	Marks	Full credit	Partial	Zero
Results on OR scale	5	Coefficients exponentiated to aORs with 95% CIs	ORs without CIs	Log-odds reported as if ORs
Complete-case reporting	2	States ~992 complete cases used	Not reported	—
Model discrimination	4	AUC/C-statistic reported (~0.71) with brief comment	Reported without comment	Absent

Earns marks: `broom::tidy(model, exponentiate = TRUE, conf.int = TRUE)`; reference levels chosen so ORs read intuitively; AUC via `pROC`.

Loses marks: linear regression on a binary outcome; outcome reversed (No as event); reading `raw_coef()` as odds ratios; dropping diabetes because the CI is wide.

6. Interpretation & Results writing — 15 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Correct clinical reading of aORs	6	Direction + magnitude right; names independent determinants	Some misreads	Reversed/incorrect throughout
Handles non-significance correctly	3	Distance described as NS / inconclusive, not "no effect"	Loosely worded	"Distance has no effect"
Results section (250–400 words, manuscript style)	4	Integrates sample flow, Table 1, model; numbers with CIs; within word count	Present but thin / off length	Missing
Precision / limitations awareness	2	Notes wide CI on diabetes (small subgroup), confounding (residence crude vs adj)	Mentions vaguely	None

Earns marks: "Diabetes was associated with over three-fold higher odds of uptake (aOR 3.56, 95% CI 1.46–9.61), though the wide CI reflects a small diabetic subgroup."

Loses marks: causal language ("urban residence causes uptake"); interpreting NS as proof of no association; reporting ORs without CIs in the prose.

7. Reproducibility & code quality — 5 marks

Sub-criterion	Marks	Full credit	Partial	Zero
Runs end-to-end from raw, clean session	2	Top-to-bottom run, no manual steps	Runs with minor fixes	Fails to run

Sub-criterion	Marks	Full credit	Partial	Zero
Relative paths / portability	1	Relative paths; no machine-specific paths	Mixed	Absolute paths (C:\Users\...)
Readability	1	Commented, sectioned, libraries loaded up top	Sparse	Unstructured
Environment captured	1	sessionInfo() / seed where needed	Partial	None

Loses marks: hard-coded absolute paths; `rm(list=ls())/setwd()` to a personal folder; packages used but never loaded.

Model answer / answer key

> Mark for **correct method + interpretation**. CIs and directions are stable; second-decimal differences across R/package versions are **acceptable**.

Expected sample flow

- Imported: **1,503** rows →
- After removing 3 duplicates: **1,500** unique patients →
- Restricted to diagnosed hypertensives (`htn_diagnosed == "Yes"`): **≈ 1,089** →
- Complete cases in the adjusted model: **≈ 992**.
- Overall treatment uptake among diagnosed: **≈ 47%**.

Canonical adjusted odds ratios (final multivariable model)

Predictor	aOR	95% CI	p	Direction
Age (per year)	1.03	1.02–1.04	<0.001	↑ uptake
Sex: Male (vs Female)	0.74	0.56–0.97	0.031	↓ uptake (female favoured)
Education (linear trend)	1.96	1.39–2.77	<0.001	↑ with higher education
Residence: Urban (vs Rural)	1.87	1.41–2.49	<0.001	↑ uptake
Diabetes: Yes	3.56	1.46–9.61	0.007	↑ uptake (largest effect, widest CI)
Family history of HTN: Yes	1.91	1.44–2.54	<0.001	↑ uptake
Health insurance: Yes	2.05	1.54–2.74	<0.001	↑ uptake
Knowledge score (per point)	1.10	1.06–1.14	<0.001	↑ uptake
Distance to facility (per km)	0.98	0.96–1.01	0.128	NS (trend ↓)

Model discrimination: AUC **≈ 0.71** (acceptable).

A correct interpretation should state:

- Independent determinants of uptake: older age, **female** sex, higher education, urban residence, diabetes, family history, health insurance, and greater hypertension knowledge.
- Diabetes** has the largest effect ($>3\times$ odds) but the **widest CI** (small subgroup) — a precision teaching point.
- Distance** is in the expected (protective-against-uptake) direction but **not significant** after adjustment — describe as inconclusive, not "no effect".
- Residence: comparing crude vs adjusted OR illustrates **mild confounding**.

Accept reasonable variation in candidate-variable lists and reference coding, provided the pre-specified core determinants are present, ORs are exponentiated with CIs, and the diagnosed cohort is used.

Common errors and mark penalties

Error	Penalty
Leaving sentinels 999 / -99 / blanks as real numbers	−4 (Component 1)
Duplicates not removed (1,503 rows analysed)	−3 (Component 1)
Impossible values (age 200, SBP 700, etc.) left in	−3 (Component 1)
Analysing all 1,500 instead of diagnosed hypertensives	−4 (Component 4) and cascade if model is also wrong
Reporting log-odds instead of exponentiated ORs	−5 (Component 5)
Outcome reversed / wrong event level	−3 (Component 5)
Linear regression on the binary outcome	−5 (Component 5)
ORs reported without 95% CIs	−2 to −3 (Components 5/6)
Interpreting a non-significant result as "no effect"	−3 (Component 6)
Causal language for a cross-sectional association	−2 (Component 6)
Absolute file paths / <code>setwd ()</code> to a personal folder	−1 (Component 7)
Script fails to run top-to-bottom	−2 (Component 7)
Table 1 / figures not exported	−3 (across Components 2 & 3)

Penalties are capped at the marks available for the relevant component — a single mistake cannot drive a component below zero.

Grade bands

Band	Marks	Descriptor
Distinction	80–100	Clean reproducible pipeline from raw data; correct cohort and sample flow; polished Table 1 and 300-dpi figures; correctly specified logistic models with exponentiated aORs and CIs; AUC reported; accurate, appropriately cautious clinical interpretation and a well-structured Results section.

Band	Marks	Descriptor
Merit	65–79	Sound end-to-end analysis with mostly correct cleaning, the right population, and a valid adjusted model on the OR scale. Minor gaps (a missed sentinel, a thin Results section, one mis-chosen test) but conclusions are correct.
Pass	50–64	Core analysis attempted and broadly correct, but with notable weaknesses — incomplete cleaning, weak Table 1/figures, partial reporting of CIs, or over-stated interpretation. Demonstrates competence but not polish.
Fail	<50	Fundamental errors: wrong population, sentinels/duplicates left in, log-odds reported as ORs, linear model on a binary outcome, or a non-reproducible/non-running submission.